

TingWei (Wade), Ou

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EDUCATION

- **National Yang Ming Chiao Tung University (NYCU)** September 2023 - Present
M.S. in Robotics Hsinchu, Taiwan
 - GPA: 3.9/4.3
- **National Yang Ming Chiao Tung University (NYCU)** September 2019 - July 2023
B.S. in Mechanical Engineering Hsinchu, Taiwan
 - GPA: 3.85/4.3

RESEARCH INTERESTS

Resource-Efficient Perception, Visual SLAM and Localization, 3D Vision for Mobile Robots and Autonomous Driving

RESEARCH EXPERIENCE

- **Autonomous Driving Department, Industrial Technology Research Institute (ITRI)** August 2026 - Present
Research Intern Hsinchu, Taiwan
 - **Project: Image-to-Point-Cloud Localization**
 - * Developing cross-modal localization methods that register camera images against LiDAR point-cloud maps for autonomous driving applications.
- **Human and Machine Lab, NYCU** September 2023 - Present
Graduate Research Student Hsinchu, Taiwan
 - **Project: Descriptor Enhancement for Visual SLAM (Desc++)**
 - * Proposed an open-source, plug-and-play descriptor enhancement module (hybrid Mamba-AFT) that enriches descriptors with spatial context in linear time, without modifying existing SLAM pipelines; released code and pretrained models for both handcrafted and learned descriptors (ORB, SIFT, SuperPoint, ALIKE).
 - * Achieved state-of-the-art matching accuracy on HPatches, nearly doubling the relative gain of the previous best method (19.0% vs. 10.2% on ORB), and reduced trajectory error across four SLAM systems on EuRoC, KITTI, and Hilti Challenge 2023.
 - * Sustained real-time performance (30 FPS, +5.3 ms added latency on an RTX 4090) with $11\times$ lower GPU memory (544 MiB vs. 6130 MiB) than a representative learned front-end; integrated drop-in with ORB-SLAM2/3, RGB-L, and MAVIS-SLAM, and demonstrated on-robot on an embedded Jetson Orin NX.
 - **Project: Autonomous UV Disinfection Robot**
 - * Proposed a hotspot-aware disinfection strategy that allocates UV dosage by contact risk rather than uniformly, reflecting that contamination concentrates on high-touch surfaces shaped by human activity.
 - * Formulated manipulator speed optimization as a linear program over a physics-based irradiance model (within 10% of measured UV-C dose), guaranteeing each surface point its risk-specified minimum dosage.
 - * Demonstrated in operational hospital wards and clinics that the strategy meets the same dosage requirements as full-coverage baselines while reducing execution time by $\sim 31\%$.
- **Networked Control Robotics Lab, NYCU** March 2020 - October 2022
Undergraduate Research Student Hsinchu, Taiwan
 - **Project: Autonomous Mobile Robot for TDK Robocon**
 - * Built a mecanum-wheeled autonomous robot from scratch as part of a student team, completing four sequential manipulation tasks (ball grasping, hoop recognition and shooting, bowling) fully autonomously within a 4-minute limit; awarded Honorable Mention.
 - * Designed the drive chassis and its control architecture spanning an onboard PC and Arduino, and implemented PID velocity control for the mecanum-wheel platform.
 - * Adopted Hector-SLAM (scan-matching) fused with triangulation-based localization to overcome unreliable wheel odometry from mecanum slippage and disturbances from dynamic bystanders near the field.

PUBLICATIONS

1. **Ting-Wei Ou**, Huang-Ting Lin, and Kuu-Young Young. Desc++: Efficient Descriptor Enhancement for Data Association in Existing Visual SLAM Systems. *Under review at IEEE Access*, 2026. [[arXiv:2607.11099](#)]
2. **Ting-Wei Ou**, Jia-Hao Jiang, Guan-Lin Huang, and Kuu-Young Young. Development of an Autonomous Mobile Robotic System for Efficient and Precise Disinfection. In *Proceedings of the IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, 2025. [[arXiv:2507.11270](#)]

CONFERENCE PRESENTATIONS

- **Desc++: Efficient Descriptor Enhancement for Data Association in Existing Visual SLAM Systems** 2026
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Pittsburgh, USA LBR
- **Development of an Autonomous Mobile Robotic System for Efficient and Precise Disinfection** 2025
IEEE International Conference on Systems, Man, and Cybernetics (SMC), Vienna, Austria Oral
- **Application of BERT-based Language Model for Visual SLAM in Feature Selection** 2024
International Conference on Advanced Robotics and Intelligent Systems (ARIS), Taipei, Taiwan LBR

OPEN SOURCE PROJECTS

- **Efficient Deployment of Learned Feature Descriptors for Visual SLAM** May 2025 - June 2025 [🌐]
 - Integrated FeatureBooster into ORB-SLAM2 to improve feature utilization, and applied model quantization to enable real-time inference on a laptop-grade GPU (RTX 2050).
 - Reduced model size by over 50%, achieving faster inference than the original paper's RTX 3090 setup while maintaining competitive matching performance.
- **ROS-Based Implementation of Multi-Camera-Inertial SLAM** January 2025 - March 2025 [🌐]
 - Developed a ROS wrapper for OpenMAVIS that runs directly on live rosbag streams, removing the offline data-conversion step required by the original release.
 - Extended evaluation to Hilti multi-camera datasets and added image compensation for robust operation in low-light environments.

WORK EXPERIENCE

- **Hong Lang Technology Co., Ltd.** [🌐] January 2023 - August 2024 Remote
Robotics Engineer
 - Led a 4-member team developing a robotic plasma-treatment system that replaces manual surface roughening—a dusty, skill-dependent process—in footwear manufacturing.
 - Designed geometry-based trajectory generation that converts the reconstructed surface into robot plasma-processing paths.
 - Reduced per-shoe processing time from 3–5 minutes (manual) to ~30 seconds in industrial demonstrations.

TEACHING AND MENTORING

- **Teaching Assistant, ECE (graduate course), NYCU – 535311: Robotics** February 2025 - June 2025
 - Supervised course projects on forward/inverse kinematics and trajectory simulation of a Stanford Arm manipulator.
- **Undergraduate Research Mentor, NYCU – Visual SLAM in Dynamic Environments** September 2024 - September 2025
 - Guided an undergraduate student in integrating YOLOv8-based dynamic object detection with ORB-SLAM2 to improve localization robustness in dynamic scenes; the student subsequently joined the lab as a graduate researcher.

LEADERSHIP EXPERIENCE

- **Team Leader, NYCU iTron Robotics Team** March 2020 - March 2021 [🌐]
 - Led a student robotics team of ~15 members, covering recruitment, technical training, and project management across annual competitions.
 - Directed the team's TDK Robocon campaign, coordinating sub-teams across mechanical, control, and software; the team earned an Honorable Mention.

HONORS AND AWARDS

- 2024–2026 Graduate Research Scholarship, NYCU Robotics
- 2025 NSTC Graduate Student International Conference Travel Grant
- 2023 Master's Program Entrance Scholarship, NYCU Robotics
- 2021 Honorable Mention Award, 26th TDK Robocon (Autonomous Robotics Group)

TECHNICAL SKILLS

- **Programming:** Python, C++, C
- **Robotics:** ROS, Visual SLAM, Visual-Inertial SLAM, Navigation, Localization, Sensor Fusion
- **Machine Learning & Computer Vision:** PyTorch, OpenCV, TensorRT
- **Development Tools:** Linux, Git, Docker, CMake, Eigen, PCL
- **Simulation & Hardware Design:** Gazebo, SolidWorks, Fusion 360, AutoCAD
- **Languages:** Chinese (Native), English (TOEFL iBT: 5.0/6.0; 100 on legacy 0–120 scale)